

Exercise 1

Selecting and Sorting

1. **SELECT ***
FROM employees;

employees						
Id	first-name	last-name	department	Salary	hire-date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesbur
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannes
6						

2. **SELECT DISTINCT** department
FROM employees;

department	
IT	
HR	
Finance	
Marketing	

3. **SELECT** first-name,
last-name
FROM employees
ORDER BY Salary **DESC**;

first-name	last-name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. **SELECT** id,
first-name,
last-name,
Salary
FROM employees
ORDER BY Salary **DESC**
LIMIT 3;

Id	first-name	last-name	Salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

Filtering with **WHERE**

5. **SELECT** id,
first-name,
last-name,
department
FROM employees
WHERE department = 'IT';

Id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6 **SELECT** id,
first-name,
last-name,
department,
Salary
FROM employees
WHERE department = 'Finance' **AND** salary > 60000;

Id	first-name	last-name	department	Salary
3	Cathy	Zulu	Finance	65 000

7. **SELECT** id,
first-name,
last-name,
department
FROM employees
WHERE department **IN** ('HR', 'marketing');

Id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. **SELECT** Id,
 first-name,
 last-name,
 department
FROM employees
WHERE department **NOT IN** ('IT');

Id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. **SELECT** Id,
 first-name,
 last-name, department
FROM employees
WHERE department **IN** ('IT', 'HR', **OR** 'Finance');

Id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. **SELECT** id,
first-name,
last-name,
department,
Salary,
City

FROM employees

WHERE department = 'IT'

AND Salary > 65000 **AND** City = 'Johannesburg'

Id	first-name	last-name	department	Salary	City
5	Eva	Naidoo	IT	72000	Johannesburg